# Technical Specification

## Adobe form ZF_LOAN_RECEIPT - book loan receipt

| Item | Value |
|---|---|
| Document | TS-20261105-library-books-loans |
| App | library-books-loans |
| Date | 05.11.2026 |
| Author | RAP Studio · AI agent |
| Request | Build an Adobe form (XDP layout) ZF_LOAN_RECEIPT for a book loan with book and borrower data, loan dates, days overdue and a late fee computed by a calculate script, a Page n of m footer and a fillable Remarks field, using a form data provider, plus a console class that reads a loan, renders the PDF with cl_fp_ads_util=>render_pdf and puts it into the print queue. |
| Status | Delivered |

## 1. Purpose and scope

A print form for the library's loans, built the SAP way: a form data provider (CDS view ZI_LoanReceipt with the service definition ZFDP_LOAN_RECEIPT), an XFA layout zf_loan_receipt.xdp with its form object, and the calling program ZCL_LOAN_RECEIPT_PRINT. The overdue days and the late fee (0.50 EUR per day) are calculated by FormCalc calculate scripts inside the form; the Remarks field is fillable in the interactive PDF.

## 2. Requirements

- Adobe form ZF_LOAN_RECEIPT with an XDP (XFA) layout
- Header data: book title, author, borrower and loan dates
- Days overdue and late fee computed by a calculate script in the form
- Footer with Page n of m
- Fillable Remarks field
- Data read through a form data provider (service definition with leading entity)
- Console class that reads a loan, renders the PDF with cl_fp_ads_util=>render_pdf and creates a print queue item

## 3. Objects

| Object | Type | File | Lines | Description |
|---|---|---|---|---|
| ZBOOK_A | Table | zbook_a.tabl.cds | 16 | |
| ZBP_R_BOOK | Class | zbp_r_book.clas.abap | 6 | |
| ZBP_R_BOOK | Class (local types) | zbp_r_book.clas.locals_imp.abap | 243 | |
| ZC_BOOK | Behavior definition | zc_book.bdef.asbdef | 24 | |
| ZC_BOOK | CDS view | zc_book.ddls.asddls | 27 | |
| ZC_BOOK | Metadata extension | zc_book.ddlx.asddlxs | 58 | |
| ZC_LOAN | CDS view | zc_loan.ddls.asddls | 29 | |
| ZC_LOAN | Metadata extension | zc_loan.ddlx.asddlxs | 48 | |
| ZCL_LIBRARY_CONSOLE | Class | zcl_library_console.clas.abap | 115 | |
| ZCL_LOAN_RECEIPT_PRINT | Class | zcl_loan_receipt_print.clas.abap | 76 | Console class (if_oo_adt_classrun): selects the open loan with the oldest due date, reads the FDP XML, renders |

| Object | Type | File | Lines | Description |
|---|---|---|---|---|
| | | | | the PDF and puts it into print queue DEFAULT |
| ZF_LOAN_RECEIPT | Print form | zf_loan_receipt.form.json | 6 | Adobe form: form object zf_loan_receipt.form.json and XFA layout zf_loan_receipt.xdp |
| ZFDP_LOAN_RECEIPT | Service definition | zfdp_loan_receipt.srvd.srvdsrv | 6 | Service definition with @ObjectModel.leadingEntity ZI_LOANRECEIPT (the form data provider) |
| ZI_BOOKLOANSTATS | CDS view | zi_bookloanstats.ddls.asddls | 13 | |
| ZI_LOANRECEIPT | CDS view | zi_loanreceipt.ddls.asddls | 26 | CDS root view entity, form data provider: loan with book title, author, ISBN, genre (via association to ZR_Book), borrower, dates and the reference date for the overdue calculation |
| ZLIBRARY | Message class | zlibrary.msag.json | 8 | |
| ZLOAN_A | Table | zloan_a.tabl.cds | 13 | |
| ZR_BOOK | Behavior definition | zr_book.bdef.asbdef | 74 | |
| ZR_BOOK | CDS view | zr_book.ddls.asddls | 28 | |
| ZR_LOAN | CDS view | zr_loan.ddls.asddls | 19 | |
| ZUI_LIBRARY | Service definition | zui_library.srvd.srvdsrv | 6 | |

# 4. Design

## Data provider

ZI_LoanReceipt selects from the loan table zloan_a and reads the book data through a to-one association to ZR_Book (path expressions). ReferenceDate is the return date, or the system date while the book is still out; the form counts the overdue days up to this date.

```
case when return_date is initial then $session.system_date else return_date end as ReferenceDate
```

## Layout

A4 portrait. The master page carries the library name, the title, a footer note with the fee rate and 'Page n of m' (floating fields with xfa.layout.page / pageCount). The body flows through the subforms Book, Borrower, Loan, Fees and RemarksBlock.

- Status (FormCalc): Returned, Overdue or On loan
- DaysOverdue (FormCalc): Max(0, Date2Num(Loan.ReferenceDate, "YYYY-MM-DD") - Date2Num(Loan.DueDate, "YYYY-MM-DD"))
- LateFee (FormCalc): Round(DaysOverdue * 0.50, 2), shown as num{z,zz9.99}
- Remarks: open multi-line text field (max. 500 characters); the form's config sets interactive output, so it becomes a PDF form field

## Output

ZCL_LOAN_RECEIPT_PRINT uses the ABAP Cloud form APIs: cl_fp_form_reader, cl_fp_fdp_services (get_keys, read_to_xml_v2 with key LOANUUID), cl_fp_ads_util=>render_pdf (PDF, pages, trace), cl_fp_ads_util=>render_4_pq and cl_print_queue_utils=>create_queue_item_by_data. Errors (cx_fp_fdp_error, cx_fp_form_reader, cx_fp_ads_util) are written to the console.

## 5. Limitations and open points

- The late fee rate (0.50 EUR per day) is a constant in the form's calculate script, not customizing data.
- The console class prints one loan (the open loan with the oldest due date); it has no selection parameter.
- Seed loans in the existing database have no member email, so the Email line is empty for them.

## 6. Quality

- Code checks (ATC, check variant Default check variant): 0 errors, 0 warnings, 5 infos (ATC Check Report).
- Tests: 5 of 5 passed (QA Test Report).